

GUIDELINES OF THE BPMN COMMUNITY ABOUT BPMN EXTENSIONS PROPOSAL

The guidelines adopted in this work were inspired by the study by Gonçalves, E., De Oliveira, M., Monteiro, I., Castro, J., and Araújo, J., 'Understanding what is important in BPMN extension proposals: the viewpoint of researchers', *Requirements Engineering Journal*, 2018. Although originally proposed in the context of the iStar language, these guidelines prove to be widely applicable to the BPMN extension scenario, since both languages share challenges related to standardisation, consistency and interoperability. Furthermore, the adoption of best practices aligned with established communities is in line with the recommendations in the *Object Management Group (OMG), Business Process Model and Notation (BPMN) – Version 2.0, 2011*, which encourages the use of systematic and well-founded approaches to the evolution of modelling languages. Consequently, it was decided to adopt these guidelines, given their relevance and potential contribution to the BPMN context.

These guidelines are listed below:

- **G1 - Preserve the language (BPMN) original syntax**
It recommends that all nodes and links of the default BPMN syntax should be maintained in the extension. In other words, the non-conservative extensions, which remove BPMN default constructs, are discouraged. So, eliminating the representation of quality, for example, is not considered a good practice;
- **G2 - Carry out consistent, complete and without conflicts extensions and follow a process/method to do them**
The BPMN extensions have been proposed in an ad-hoc fashion manner. This fact has a negative impact on the proposal and many problems of inconsistencies, incompleteness and conflicts have been occurring. Therefore, this guideline highlights the importance of to follow a process (PRISE) to avoid inconsistencies, incompleteness and conflicts;
- **G3 - Perform a literature review, participation of domain experts and BPMN experts and model systems of application area before extending**
The literature review can contribute with the understanding of the domain and to set clear definitions to the concepts of this domain. The participation of domain experts and BPMN can contribute to avoid errors during BPMN extensions proposal;
- **G4 - Describe a clear definition of the extension concepts**
Concepts which are not described cannot be understood by those who intend to use the extension;
- **G5 - Propose concrete and abstract syntax of the extension**
The metamodel and well-formedness rules are important to a modelling language and their extensions, once they represent what is allowed and what is forbidden in its usage. Therefore, both syntaxes (Abstract and concrete) should be considered during the extension proposition;
- **G6 - Check consistency between abstract and concrete syntaxes**
It is important to consider the Checking consistency between abstract and concrete syntaxes, once the syntaxes can be inconsistent. The inconsistency can be described by the representation of a new construct in only one of them, for example including a new graphical representation in concrete syntax, without adding a new metaclass to represent it in the extension metamodel;

- **G7 - Relate concepts introduced by the extensions with the BPMN concepts** *The new concepts introduced by the extension can be related to the BPMN concepts, specialising the BPMN constructs or be connected by BPMN relationships or other relationships*
- **G8 - Define extensions with a smallest possible number of modifications and new representations in order not to complicate the use of the modelling language (BPMN)** *The extension should minimise the number of modifications and new representations in BPMN as much as possible. It is important that the BPMN extension does not introduce a great number of constructs, making the usage of the BPMN extension hard. Otherwise, it can mischaracterize BPMN and worsen its scalability;*
- **G9 - Propose careful and simple graphical representations, able to be drawn on paper without a tool** *The proposal of new constructs should be simple enough for people to hand draw and do it on the whiteboard and scrap it off and make changes to it very quickly. They should maintain the principle of the symbols of BPMN, which are simple and easy to draw. If possible, in order to choose an appropriate graphical representation, try to perform an experiment such as the one suggested by CAIRE et al. in P. Caire, N. Genon, P. Heymans, D. Moody. "Visual notation design 2.0: Towards user comprehensible requirements engineering notations". 21st IEEE International Requirements Engineering Conference (RE), 2013.*
- **G10 – Define quality requirements based on ISO/IEC 25010:2011** *It is recommended that extension requirements be defined based on the quality model proposed by ISO/IEC 25010:2011. The extension proposal should identify the most relevant quality attributes (e.g. usability, performance, reliability) and specify them in a structured and systematic manner. This guideline ensures clarity, consistency and alignment with internationally recognised quality standards.*
- **G11 - Assess the suitability of BPMN before proposing the extension** *This guideline highlights the importance of verifying whether BPMN is an appropriate modelling language for the intended extension. A structured checklist (e.g., BPMN Conformity Checklist) should be used to assess conceptual alignment and adherence to the business process domain. If BPMN is not suitable, the extension should not proceed and another modelling language should be considered.*
- **G12 - Provide a structural representation aligned with the BPMN metamodel** *It is recommended that the extension include a structural specification, such as a class diagram, describing classes, attributes, relationships and specialisations. This representation must be consistent with the BPMN metamodel and follow the Object Management Group's guidelines to ensure standardisation and interoperability.*
- **G13 - Systematise the conceptual and structural characterisation of the extension** *Recommends organising the mapping between extended elements and BPMN constructs in a structured manner. This includes documenting equivalences, relationships and distinctions, enabling a clearer understanding of how the extension integrates with BPMN.*
- **G14 - Provide an XML representation compliant with OMG standards** *The extension must include an XML schema specification aligned with the BPMN standards defined by the Object Management Group. This ensures that the extension can be formally represented, exchanged and integrated with tools that support BPMN XML.*
- **G15 - Consider tool support for the proposed extension** *This guideline highlights the importance of assessing whether the extension requires tool support (e.g. modelling, validation, execution). If so, the extension should be implemented or*

adapted within a modelling tool to ensure practical applicability and usability.

- **G16 - Ensure the implementation of the extension in modelling tools**

It is recommended to incorporate the new constructs into modelling tools (e.g. via SVG or equivalent representations), ensuring visual and functional support. This facilitates the practical adoption and validation of the extension in real-world modelling scenarios.